

SDLC Approach Comparison with Model

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System Description: The system will be a minimalist workout tracker app. It will have an easy-to-use, clean UI. Users will be able to create and manage new workouts consisting of several exercises. For each gym session, users will select and start a workout from the app. For each exercise, they will get a performance history as reference. The users will then perform the exercise and track the new results. At the end of each workout, they will see a comprehensive review and comparison with past workouts. The app should operate entirely locally and offline, to allow smooth and responsive feedback.

Predictive Approach: The design and requirements are determined before implementation starts. All features will be planned ahead of time. This includes UI, logic, databases, workout review, etc.

Adaptive Approach: Start with core, basic features and adapt based on changing user requirements, technology, or priorities. Implement a rough UI, database, and review system and then change or add features based on user feedback. Scope and objectives will become incrementally clearer as development progresses.

Iterative Approach: Define general requirements and separate them into iterations. For example, start with core features such as rough UI, save and load workouts (database), exercise performance tracking, and basic history display. Then re-iterate by refining UI, improving database, and adding proper history review with graphs. Finally, iterate once again by refining, fixing bugs, and making sure to have met all requirements.

Recommended Approach & Justification: For this system, the best approach would be the iterative approach with hints of the adaptive approach. The scope and requirements of the entire mobile app are difficult to fully determine before development starts for the predictive approach. Designing everything and building it in a single pass is not feasible or realistic. Furthermore, while there are risks such as incorrect tracking, which may cause injury to the user, it is not high-risk or uncertain enough to warrant a purely adaptive approach. The iterative approach is best because it allows for basic functionality very quickly while still delivering a polished and finished product after all iterations. It also allows for testing between iterations, which may improve the general quality. However, due to personal inexperience with mobile development, some level of adaptive approach may be necessary. It would also allow for easy feature expansion later, which is a realistic possibility for a gym-related app. Therefore, following the spiral model would work best, as it encompasses both the iterative and adaptive approaches. It allows for risk analysis and some uncertainty in early development, where technical or usability issues may arise.

